experts, who had tried to be too cunning by submitting subtly nasty strategies; while even those who had submitted nice strategies had not dared anything so forgiving as Tit for Two Tats.

Axelrod announced a second tournament. He received 62 entries and again added Random, making 63 in all. This time, the exact number of moves per game was not fixed at 200 but was left open, for a good reason that I shall come to later. We can still express scores as a percentage of the 'benchmark', or 'always cooperate' score, even though that benchmark needs more complicated calculation and is no longer a fixed 600 points.

Programmers in the second tournament had all been provided with the results of the first, including Axelrod's analysis of why Tit for Tat and other nice and forgiving strategies had done so well. It was only to be expected that the contestants would take note of this background information, in one way or another. In fact, they split into two schools of thought. Some reasoned that niceness and forgivingness were evidently winning qualities, and they accordingly submitted nice, forgiving strategies. John Maynard Smith went so far as to submit the super-forgiving Tit for Two Tats. The other school of thought reasoned that lots of their colleagues, having read Axelrod's analysis, would now submit nice, forgiving strategies. They therefore submitted nasty strategies, trying to exploit these anticipated softies!

But once again nastiness didn't pay. Once again, Tit for Tat, submitted by Anatol Rapoport, was the winner, and it scored a massive 96 per cent of the benchmark score. And again nice strategies, in general, did better than nasty ones. All but one of the top 15 strategies were nice, and all but one of the bottom 15 were nasty. But although the saintly Tit for Two Tats would have won the first tournament if it had been submitted, it did not win the second. This was because the field now included more subtle nasty strategies capable of preying ruthlessly upon such an out-and-out softy.

This underlines an important point about these tournaments. Success for a strategy depends upon which other strategies happen to be submitted. This is the only way to account for the difference between the second tournament, in which Tit for Two Tats was ranked well down the list, and the first tournament, which Tit for Two Tats would have won. But, as I said before, this is not a book about the ingenuity of computer programmers. Is there an objective way in which we can judge which is